

KARTHIKEYAN SIVARAJ

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PROFESSIONAL SUMMARY

Aspiring Data Scientist with hands-on machine learning and BI project experience: tested feature engineering, cost-optimized deployment, and drift monitoring, built and deployed independently after formal data science training in 2025. Prior 5 years in design brought strong data-storytelling and stakeholder-facing communication skills, now applied to dashboards and executive-ready analytics. Effective at transforming complex information into clear visual narratives and business insights that drive decisions.

CORE COMPETENCIES

Python | SQL | Machine Learning | Predictive Analytics | Data Cleaning | Data Wrangling | EDA | Feature Engineering | Statistics | Hypothesis Testing | Regression | Classification | Clustering | Model Evaluation | Cross Validation | Power BI | Tableau | Excel | Data Visualization | Dashboard Development | Business Intelligence | ETL | KPI Reporting | Data Storytelling | Stakeholder Management | Problem Solving

TECHNICAL SKILLS

Programming and Data: Python (pandas, NumPy, scikit-learn, LightGBM, XGBoost, Optuna, SHAP, statsmodels, Seaborn, Matplotlib), SQL (PostgreSQL, stored procedures, triggers, query optimization).

Machine Learning: Regression, Classification, Clustering, imbalanced data (PR-AUC, class weighting), feature engineering, feature selection, leakage prevention, cross validation, hyperparameter tuning, model explainability, drift monitoring, A/B test design.

Engineering and MLOps: FastAPI, REST API, Streamlit, Docker, GitHub Actions (CI/CD), pytest, Git, GitHub, MLflow, Jupyter Notebook, automation.

Power BI and Visualization: Power BI (DAX, star schema), Tableau, Excel (Power Query, PivotTables), Plotly; dashboard development, KPI reporting, data storytelling.

Statistics: Hypothesis testing (z-test, chi-square, ANOVA), probability, confidence intervals, correlation analysis, cost-benefit and sensitivity analysis.

PROJECTS

ShipGuard - E-commerce Return-to-Origin (Failed Delivery) Prediction Platform | Python, LightGBM, PostgreSQL, FastAPI, Docker, SHAP | [GitHub: github.com/karthikeyan-sivaraj/shipguard-rto-intelligence](https://github.com/karthikeyan-sivaraj/shipguard-rto-intelligence)

- Built an end-to-end machine learning system predicting Cash-on-Delivery return-to-origin risk at order placement on a 150,000-order relational dataset (8 PostgreSQL tables); engineered leakage-tested, point-in-time customer-history features validated by unit tests.
- Compared 6 model families on a time-based split and selected the operating threshold by cost optimization instead of F1, reaching **92% recall and a projected Rs 5 lakh per year in net savings**.
- Deployed the model as a FastAPI service returning SHAP explanation codes; shipped a Streamlit operations console, an executive dashboard, a Docker Compose stack, a CI pipeline that retrains on sample data before merge, and drift monitoring (PSI) that caught a planted courier degradation.

CareSlot - Hospital No-Show Prediction and Overbooking Optimization | Python, scikit-learn, PostgreSQL, FastAPI, Streamlit, Docker | [GitHub: github.com/karthikeyan-sivaraj/hospital-noshow-overbooking](https://github.com/karthikeyan-sivaraj/hospital-noshow-overbooking)

- Built a clinic no-show prediction and daily overbooking optimization platform on a 150,000-appointment dataset (12 clinics, 99 providers); stage-aware leakage rules keep reminder response valid for day-before scoring but blocked for booking-time models (config-gated and test-enforced).
- Designed an **exact Poisson-binomial overbooking optimizer** (dynamic-programming convolution, unit-tested against the closed-form Binomial to $1e-12$) that decides waitlist backfills per clinic-day using newsvendor cost logic (idle slot Rs 900 vs overflow Rs 350).
- On a held-out quarter with realized outcomes, cut cost per clinic-day from **Rs 6,298 to Rs 2,392, capturing 74% of a perfect-foresight benchmark** (projected Rs 1.7 crore per year); the model achieved PR-AUC 0.31 vs a 0.17 base rate, consistent with published no-show research.

Customer Support Analytics Dashboard | Tableau, Excel, Star Schema

- Modeled 2,000 support tickets into a star schema (fact + 6 dimensions across agents, 5 channels, 4 regions) and built an interactive Tableau dashboard tracking 10+ KPIs: **89.3% resolution rate, 6.6-hr avg first response, 24.5-hr avg resolution, 20.2% repeat-contact rate.**
- Surfaced SLA compliance gaps (only **55% of tickets met response SLA and 62% met resolution SLA**, with Email the weakest channel at 59% vs Live Chat at 66%) and recommended channel rebalancing and priority-based routing, supported by escalation and CSAT analysis.

Patient Experience Analytics Dashboard | Power BI, DAX

- Cleaned and modeled 2,000 patient-feedback records (44 fields; 5 departments, 10 doctors) into a Power BI star schema with documented DAX measures; the dashboard tracks **satisfaction 3.12 of 5, NPS -50 (60% detractors), 32.6-minute average wait, and 13.5% service-recovery flags**, with department, doctor, and age-band drill-downs.
- Identified wait time as the leading dissatisfaction driver (**correlation -0.31** with satisfaction; General Medicine worst at 34 minutes vs Cardiology at 31 minutes) and proposed queue triage and follow-up workflow improvements in an executive insights report.

Zomato Restaurant Market Intelligence | Excel, Power Query, PivotTables

- Analyzed 3,000 restaurants across 5 cities and 4 zones in Excel using Power Query cleaning, PivotTable analysis, and KPI sheets covering ratings (average 2.98), pricing (average cost for two Rs 1,095), cuisines, delivery times, order volumes, and revenue.
- Quantified digital adoption (**51% online-delivery penetration**, with delivery-enabled outlets rating higher at 3.01 vs 2.95) and delivered a zone and cuisine opportunity matrix with an executive action plan for expansion targeting.

WORK EXPERIENCE

Junior Graphic Designer | Madras Global, Chennai | Mar 2019 - Oct 2021

- Designed and delivered 200+ static and animated advertising creatives for clients across various industries, maintaining high standards of creativity and brand alignment.
- Developed digital advertising content for web, social media, and marketing campaigns, driving audience engagement and strengthening brand presence.

Creative Designer | Kumbakonam Coffee, Chennai | May 2022 - Feb 2025

- Led and managed the complete creative design function of the company, overseeing branding, marketing collateral, advertising campaigns, packaging, and visual communication initiatives.
- Designed compelling product packaging and cover designs for multiple product lines, enhancing brand appeal and contributing to increased customer interest and product demand.

EDUCATION

Bachelor of Engineering, Computer Science and Engineering

Panimalar Institute of Technology, Chennai | 2014 - 2018

CERTIFICATION

Data Science - Python, Machine Learning, SQL, Statistics, BI

Besant Technologies, Chennai | Mar 2025 - Dec 2025